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T7 — Genetics, Populations, Evolution & Ecosystems

7.1 Inheritance

Allele: An alternative form of a gene, occupying the same locus on homologous chromosomes. Alleles differ in their DNA base sequence and may code for different versions of the same polypeptide.
Genotype: The combination of alleles an organism carries for a particular gene (or genes).
Phenotype: The observable characteristics of an organism, resulting from the interaction between its genotype and environmental influences.

Key Genetic Terms

Term	Definition
Homozygous	Two identical alleles at a locus (e.g. AA or aa)
Heterozygous	Two different alleles at a locus (e.g. Aa)
Dominant	Allele expressed in both homozygous (AA) and heterozygous (Aa) genotype. Only one copy needed for expression.
Recessive	Allele only expressed in the homozygous state (aa). Masked by the dominant allele in heterozygotes.
Carrier	A heterozygous individual for a recessive condition — phenotypically unaffected but can pass the recessive allele to offspring.

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Test cross	Crossing an individual of dominant phenotype with a homozygous recessive (aa) to determine if the unknown genotype is AA or Aa.

Monohybrid Inheritance

Monohybrid crosses involve a single gene with two alleles. The classic cross between two heterozygotes:

Example: Aa x Aa (e.g. tongue rolling ability)

	A	a
A	AA	Aa
a	Aa	aa

Genotypic ratio: 1 AA : 2 Aa : 1 aa
Phenotypic ratio: 3 dominant : 1 recessive

> **Exam Tip:** When writing genetic crosses in exams, always include: (1) a key defining the alleles (e.g. A = dominant allele for..., a = recessive allele for...); (2) parental phenotypes and genotypes; (3) gametes produced; (4) a Punnett square; (5) offspring genotypic and phenotypic ratios. Show your working clearly — marks are given for method, not just the answer.

Dihybrid Inheritance

Dihybrid crosses involve two genes on **different chromosomes** (so they are independently assorted during meiosis). A cross between two double heterozygotes (AaBb x AaBb) produces a **9:3:3:1** phenotypic ratio:

- 9 = both dominant (A_B_)
- 3 = first dominant, second recessive (A_bb)
- 3 = first recessive, second dominant (aaB_)
- 1 = both recessive (aabb)

If two genes are on the same chromosome, they are **linked** and tend to be inherited together (they do not assort independently). Linkage produces ratios that deviate from the expected 9:3:3:1. Crossing over during

meiosis can break linkage, producing recombinant phenotypes at a lower frequency.

Codominance

Codominance: A pattern of inheritance where both alleles are expressed simultaneously in the heterozygote — neither is dominant over the other. The heterozygote shows a phenotype distinct from both homozygotes but displaying features of both.
Convention: codominant alleles are written as superscripts, e.g. C^RC^W.

Example — Sickle cell trait: Individuals with genotype Hb^AHb^S produce both normal and sickle haemoglobin. They have mild sickle cell trait but also resistance to malaria (heterozygote advantage).

Multiple Alleles

Some genes have more than two alleles in a population (though an individual still carries only two). The classic example is the **ABO blood group system**: three alleles I^A, I^B, and I^O.

- I^A and I^B are **codominant** with each other (both expressed in I^AI^B = blood group AB)
- Both I^A and I^B are **dominant** over I^O
- I^O is **recessive**

Blood Group	Possible Genotypes	Antigens on RBC Surface	Antibodies in Plasma
A	I^AI^A or I^AI^O	A antigen	Anti-B
B	I^BI^B or I^BI^O	B antigen	Anti-A
AB	I^AI^B	A and B antigens	Neither (universal recipient)
O	I^OI^O	Neither antigen	Anti-A and Anti-B (universal donor)

Sex Linkage

Sex-linked gene: A gene located on the sex chromosomes, usually the **X chromosome** (X-linked). Since males have only one X chromosome (XY), they are **hemizygous** for X-linked genes — a single

recessive allele will be expressed because there is no second X to carry a dominant allele. This is why X-linked recessive conditions are much more common in males.

Examples of X-linked recessive conditions:

- **Haemophilia A:** Deficiency of clotting factor VIII. Affected males (X^hY) bleed excessively. Females can be carriers (X^HX^h) or affected (X^hX^h, very rare).
- **Red-green colour blindness:** Defective cone pigment genes on the X chromosome. ~8% of males affected vs ~0.5% of females.
- **Duchenne muscular dystrophy:** Defective dystrophin gene on X chromosome.

Cross example: Carrier female (X^HX^h) x Normal male (X^HY):

	X^H	Y
X^H	X^HX^H (normal female)	X^HY (normal male)
X^h	X^HX^h (carrier female)	X^hY (affected male)

Result: 50% of sons affected; 50% of daughters are carriers. No daughters show the condition. No sons are carriers (they either have or do not have the condition).

Autosomal Linkage

When two genes are located on the **same autosome** (non-sex chromosome), they are physically linked and tend to be inherited together. This produces offspring ratios that deviate from the expected independent assortment ratios. The closer together two genes are on a chromosome, the less likely they are to be separated by crossing over, and the fewer recombinant offspring are produced.

Epistasis

Epistasis: The interaction between genes at different loci, where one gene masks or modifies the expression of another gene. The masking gene is the **epistatic gene**; the masked gene is the **hypostatic gene**. This produces modified dihybrid ratios.

Type	Modified Ratio (from 9:3:3:1)	Explanation	Example
Recessive epistasis	9:3:4	Homozygous recessive genotype at one locus masks expression of the other gene entirely	Labrador coat colour: ee genotype = yellow, regardless of B/b genotype. EE/Ee + BB/Bb = black; EE/Ee + bb = chocolate
Dominant epistasis	12:3:1	A dominant allele at one locus masks expression of the other gene	Squash fruit colour: W_ = white (masks Y/y gene). wwY_ = yellow; wwyy = green
Complementary genes	9:7	Both genes must have at least one dominant allele for the final phenotype to appear	Flower colour: functional enzyme needed from each gene (A and B). Only A_B_ = coloured; all others = white

Exam Tip: When you see unexpected ratios in a dihybrid cross (anything other than 9:3:3:1), consider epistasis. The modified ratios still add up to 16 (the total number of possible genotype combinations from a dihybrid cross). Identify which categories have been merged and this tells you which type of epistasis is occurring.

The Chi-Squared (χ^2) Test

Chi-squared test: A statistical test used to determine whether there is a significant difference between observed and expected results. It tests the **null hypothesis** (H_0): there is no significant difference between observed and expected frequencies — any difference is due to chance.

$$\chi^2 = \Sigma (O - E)^2 / E$$

Where O = observed frequency, E = expected frequency for each category.

Steps for the Chi-Squared Test

1. **State the null hypothesis:** "There is no significant difference between the observed and expected ratios."
2. **Calculate expected values** from the predicted genetic ratio (e.g. 3:1 predicts 75% and 25% of total).
3. **Calculate χ^2** using the formula for each category, then sum.
4. **Determine degrees of freedom:** df = number of phenotypic categories - 1.
5. **Compare the calculated χ^2 with the critical value** from a table at **p = 0.05** (5% significance level).
6. If χ^2 **<** critical value: **accept H_0** — the difference is not significant; results are consistent with the expected ratio (any difference is likely due to chance).
7. If χ^2 **>** critical value: **reject H_0** — the difference IS significant; results do not fit the expected ratio (some other factor is affecting the results).

Exam Tip: The p = 0.05 threshold means there is a 5% probability (1 in 20 chance) that the difference between observed and expected is due to chance alone. If your calculated χ^2 exceeds the critical value, there is less than a 5% chance the difference is due to chance, so you conclude it is significant. Always state both the null hypothesis AND your conclusion clearly. The chi-squared test requires raw count data (not percentages) and expected values of at least 5 in each category.

Worked Example

In a monohybrid cross (Aa x Aa), you expect a 3:1 ratio. From 120 offspring, you observe 82 dominant and 38 recessive.

- Expected: 90 dominant (120 x 3/4) and 30 recessive (120 x 1/4)
- $\chi^2 = (82-90)^2/90 + (38-30)^2/30 = 64/90 + 64/30 = 0.711 + 2.133 = 2.844$
- df = 2 - 1 = 1. Critical value at p=0.05, df=1 is 3.841
- 2.844 < 3.841, so accept H_0. The results are consistent with a 3:1 ratio.

7.2 Populations

Hardy-Weinberg Principle

Hardy-Weinberg Principle: In a large, randomly mating population with no selection, mutation, migration, or genetic drift, allele and genotype frequencies remain constant from generation to generation (the population is in **genetic equilibrium**).

$$p + q = 1 \quad \text{(allele frequencies: } p = \text{dominant allele frequency, } q = \text{recessive allele frequency)}$$

$$p^2 + 2pq + q^2 = 1 \quad \text{(genotype frequencies: } p^2 = \text{homozygous dominant, } 2pq = \text{heterozygous, } q^2 = \text{homozygous recessive)}$$

Hardy-Weinberg: How to Solve Problems

Step 1
Identify q^2
= frequency of recessive phenotype

→

Step 2
$q = \sqrt{q^2}$
$p = 1 - q$

→

Step 3
Carrier frequency
= 2pq

Example: Cystic fibrosis affects 1 in 2500 people
$q^2 = 1/2500 = 0.0004 \rightarrow q = 0.02 \rightarrow p = 0.98$
Carrier frequency = 2pq = 2 x 0.98 x 0.02 = 0.0392 ≈ 1 in 25 people

Conditions for Hardy-Weinberg Equilibrium

These conditions are rarely met perfectly in nature, but the model provides a useful baseline for detecting when evolution IS occurring:

- **Large population size** (no genetic drift — random fluctuations in small populations)
- **Random mating** (no mate selection based on genotype/phenotype)
- **No natural selection** (all genotypes are equally fit)
- **No mutation** (no new alleles being created)
- **No migration** (no gene flow in or out of the population)

If allele frequencies in a population change over time (deviating from Hardy-Weinberg predictions), this indicates that **evolution is occurring** — one or more of the conditions is being violated.

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7.3 Evolution and Natural Selection

Evolution: A change in allele frequency in a population over successive generations. It is driven by natural selection, genetic drift, mutation, and gene flow.

Darwin's Theory of Natural Selection

1. **Variation exists** within a population due to genetic differences (caused by mutation, meiosis, and sexual reproduction).
2. **Overproduction:** Organisms produce more offspring than the environment can support.
3. **Struggle for existence:** There is competition for limited resources (food, territory, mates) — **intraspecific competition**.
4. **Differential survival:** Individuals with alleles conferring advantageous phenotypes for the current environment are more likely to survive (they have greater **fitness**).
5. **Differential reproduction:** Survivors reproduce and pass on their advantageous alleles to offspring at a higher frequency.
6. **Cumulative change:** Over many generations, the frequency of advantageous alleles increases in the population — the population **evolves**.

Exam Tip: When describing natural selection in exam answers: (1) start with variation existing within a population due to mutation; (2) mention a specific selection pressure; (3) explain which phenotype has a survival/reproductive advantage; (4) state that advantageous alleles are passed to offspring; (5) conclude that allele frequency changes over generations. Never say organisms "choose" to evolve or "try" to adapt — evolution is not directed or intentional.

Types of Selection

Type	Description	Effect on Variation	Example
Directional	Individuals at one extreme of the phenotype range have higher fitness; the mean	Reduces variation; shifts allele frequency toward the favoured extreme	Antibiotic resistance in bacteria — resistant individuals survive and increase in frequency;

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	phenotype shifts in one direction		industrial melanism in peppered moths
Stabilising	Individuals with the intermediate (average) phenotype are favoured; both extremes are selected against	Reduces variation; maintains the mean; narrows the distribution	Human birth weight — very low or very high birth weights have higher mortality; clutch size in birds
Disruptive	Individuals at both extremes of the range are favoured over intermediates	Increases variation; can produce a bimodal distribution; may lead to speciation	Beak size in seed-cracking finches — large beaks for large seeds, small beaks for small seeds; intermediate beaks are less efficient at either

Speciation

Species: A group of organisms that can interbreed to produce fertile offspring and are reproductively isolated from other species. (Note: this biological species concept has limitations — it cannot apply to asexual organisms, fossils, or organisms that hybridise.)
Speciation: The formation of new species from an existing species, when populations become reproductively isolated and diverge genetically to the point where they can no longer interbreed to produce fertile offspring.

Allopatric Speciation

1. A population is geographically separated by a physical barrier (mountain range, river, ocean, glacier, human development).
2. The two sub-populations experience **different selection pressures** and **different mutations** occur randomly in each.
3. **Genetic drift** (especially if populations are small) further increases divergence.
4. No gene flow between the populations (they cannot interbreed because of the barrier).
5. Over many generations, the allele frequencies diverge so much that the populations develop **reproductive isolation** — they can no longer interbreed successfully even if the barrier is removed.
6. Two new species have formed.

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Example: Darwin's finches on the Galapagos Islands — ancestral finch population colonised different islands, each with different food sources. Different beak shapes evolved through natural selection on each island, eventually producing ~18 different species.

Sympatric Speciation

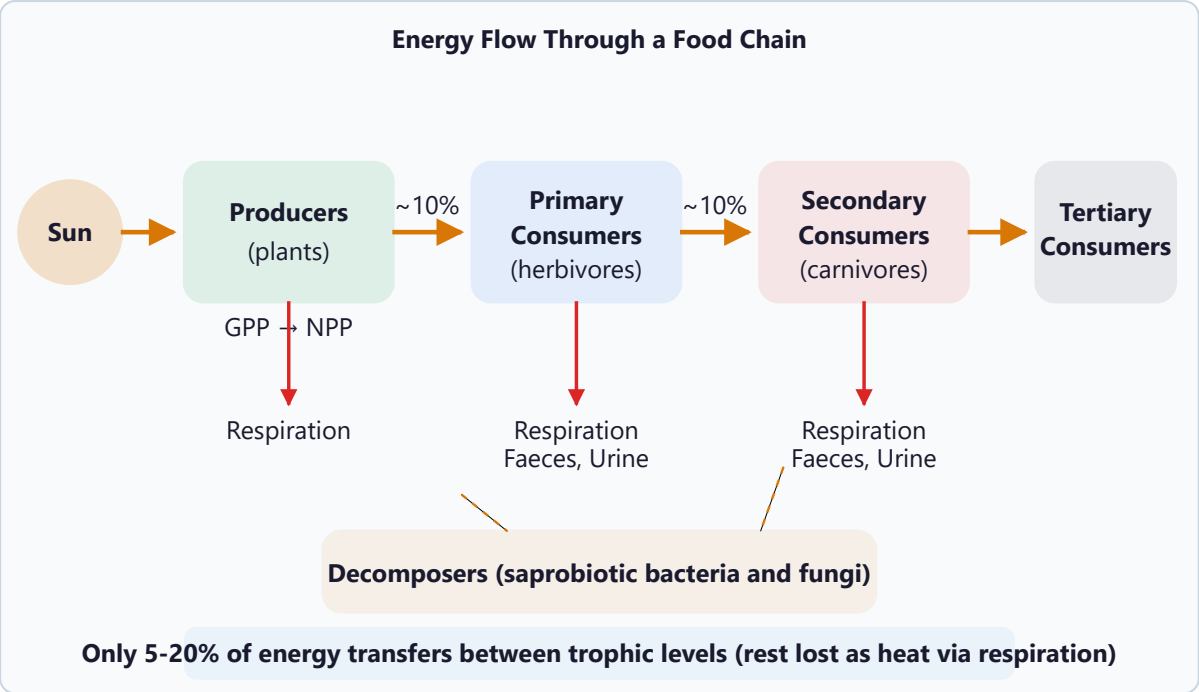
Speciation without geographic isolation. This can occur through:

- **Polyploidy** (especially in plants): A mutation error during cell division produces offspring with extra sets of chromosomes (e.g. 4n instead of 2n). These cannot breed with the original 2n population (meiosis fails to produce viable gametes when crossed). Common in plants — up to 70% of flowering plant species are polyploid. Many crop species are polyploid (wheat is 6n, strawberry is 8n).
- **Ecological niche differentiation:** Populations exploit different food sources or habitats within the same area, reducing gene flow between them.
- **Temporal reproductive isolation:** Populations breed at different times (different seasons, different times of day).
- **Behavioural isolation:** Different courtship behaviours prevent interbreeding.

7.4 Ecosystems

Ecosystem: A community of living organisms (biotic) interacting with each other and with the non-living (abiotic) components of their environment in a defined area.
Population: All the organisms of one species in a habitat at a given time.
Community: All the populations of different species living and interacting in a particular area at a given time.
Niche: The role of a species within its ecosystem — including its habitat, diet, behaviour, and all its biotic and abiotic interactions. No two species can occupy exactly the same niche in the same habitat (competitive exclusion principle).

Energy Flow Through Food Chains



Ecological Succession

Succession: The progressive, directional change in the species composition of a community over time, leading towards a stable **climax community**.

Type	Starting Point	Example
Primary succession	Bare, previously uncolonised substrate with no soil (bare rock, new volcanic island, sand dune, glacial retreat)	Lichens colonise bare rock → mosses → grasses → shrubs → trees (deciduous woodland)
Secondary succession	Previously colonised substrate where soil already exists but vegetation has been removed (after fire, flooding, deforestation, farming)	Faster than primary because soil nutrients and seed bank already exist

Stages of Succession

1. **Pioneer community:** Pioneer species (e.g. lichens, mosses, marram grass on sand dunes) colonise the bare substrate. They are adapted to harsh conditions (drought-tolerant, low nutrient requirements, rapid reproduction). They modify the environment: weathering rock, trapping organic matter, beginning soil formation, providing shade and shelter.

2. **Intermediate (seral) stages:** As soil develops and conditions improve, new species arrive that outcompete the pioneers. Each seral community modifies the environment further (deeper soil, more nutrients, different microclimate), facilitating the next stage.
3. **Climax community:** A stable, self-sustaining community in equilibrium with the prevailing climate and environment. In the UK, the climatic climax is typically **temperate deciduous woodland** (dominated by oak, ash, beech). Species diversity is typically highest at climax. The community remains stable as long as conditions do not change.

Changes during succession:

- Increasing **biomass** and **productivity**
- Increasing **species diversity** and more complex food webs
- Soil becomes **deeper**, more nutrient-rich, with greater water retention
- More **stratification** (vertical layers in the community — ground, field, shrub, canopy layers)
- Abiotic conditions become **less extreme** (more sheltered, more humid, less wind)
- **Nutrient cycling** becomes more efficient (closed cycling in climax vs open in pioneer)

Conservation and Management of Succession

Sometimes conservation goals require maintaining a particular **seral stage** (plagioclimax) rather than allowing succession to climax. For example, heathland, grassland, and coppiced woodland support specific species that would be lost if succession proceeded to dense woodland.

Management techniques:

- **Grazing** — livestock prevent succession to woodland by eating tree seedlings
- **Mowing / cutting** — maintains grassland
- **Controlled burning** — removes woody plants, promotes grass and heather regeneration
- **Coppicing** — cutting trees at base to encourage regrowth from stumps; creates varied age structure and light levels, supporting diverse species

Reasons for Conservation

Category	Arguments

Ethical	All species have intrinsic value and a right to exist, regardless of human use. We have a moral responsibility not to cause extinction.
Ecological	Maintaining ecosystem stability, resilience, and function. Maintaining food webs and nutrient cycling. Keystone species removal can cause ecosystem collapse.
Economic	Genetic resources for agriculture, medicine, and industry (e.g. plant-derived drugs). Tourism and recreation (ecotourism). Ecosystem services (pollination, water purification, flood prevention, carbon sequestration).
Aesthetic	Natural beauty, cultural heritage, spiritual value. Mental health benefits of natural environments.

Conservation methods:

- **In situ conservation:** Protecting species in their natural habitat — nature reserves, national parks, SSSIs (Sites of Special Scientific Interest), marine protected areas, wildlife corridors
- **Ex situ conservation:** Protecting species outside their natural habitat — zoos, botanical gardens, seed banks (e.g. Svalbard Global Seed Vault), captive breeding programmes, cryopreservation of gametes

Exam Tip: In extended response questions on conservation, always include specific examples and link arguments to economic, ethical, and ecological reasons. Discuss both in situ and ex situ methods, their advantages and limitations. For example, captive breeding programmes (e.g. Arabian oryx, California condor) can save species from extinction but face challenges: inbreeding depression (reduced genetic diversity), loss of natural behaviours, difficulty reintroducing to the wild, and high cost. Seed banks preserve genetic diversity cheaply but cannot conserve animal species or complex ecological relationships.

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